

TABLE I
Power Factor Multiplier

Original Power Factor	Desired Power Factor					
	0.800	0.850	0.900	0.950	1.000	
0.50	0.982	1.112	1.248	1.403	1.732	
0.55	0.768	0.899	1.034	1.190	1.518	
0.60	0.583	0.714	0.849	1.005	1.333	
0.65	0.419	0.549	0.685	0.840	1.169	
0.70	0.270	0.400	0.536	0.692	1.020	
0.75	0.132	0.262	0.398	0.553	0.882	
0.80	0.000	0.130	0.266	0.421	0.750	
0.85	—	0.000	0.135	0.291	0.620	
0.90	—	—	0.000	0.156	0.484	
0.95	—	—	—	0.000	0.329	
1.00	—	—	—	—	0.000	

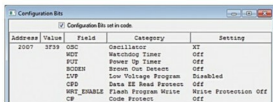


Fig. 2: Configuration bits

By improving the power factor you can save money on your electricity bill and also derive the following benefits:

1. Reduction of heating losses in transformers and distribution equipment
2. Longer equipment life
3. Increase in the capacity of your existing system and equipment.

While power factor correction is required for efficient use of electrical power, over correction of power factor is not recommended. In this project, over correction is not considered; we have considered power factors between 60 per cent and 90 per cent only due to inductive loads.

Circuit and working

Fig. 1 shows the circuit of a microcontroller-based power factor corrector. The circuit is built around PIC16F877A microcontroller (IC1), 230V AC primary to 9V, 300mA secondary transformer (X1), current transformer (X2), three relays (RL1-RL3), a 16x2 LCD display (LCD1) and a few other components.

Microcontroller PIC16F877A. Microcontroller PIC16F877A is the heart

of the circuit. It is used to detect the phase difference between voltage and current in the AC mains supply line. It also connects the power factor correction (PFC) capacitor across the inductive load through a relay as explained below.

IC1 is a low-power, high-performance, CMOS 8-bit microcontroller. Its main features are 8kB Flash memory, 256-byte EEPROM, 368-byte RAM, 33 input/output (I/O) pins, 10-bit 8-channel analogue-to-digital converter (ADC), three timers, a watchdog timer with its own on-chip oscillator for reliable operation and synchronous I²C interface.

IC1's port pins RB0 through RB7 are connected to D0 through D7 of LCD1.

Port pins RD5, RD6 and RD7 of IC1 are connected to control pins register select (RS), read/write (R/W) and enable (EN) of LCD1, respectively. Port pins RC4 through RC6 are used to control relays RL1 through RL3, respectively.

In this project, three different PFC capacitors are used to connect across the inductive load through RL1, RL2 and RL3 relays. If power factor is between 60 per cent and 70 per cent, RL1 is energized. RL2 is energized if power factor is over 70 per cent but within 80 per cent and RL3 for power factor above 80 per cent but within 90 per cent.

When port pin RC4 of IC1 goes high, transistor T3 conducts and relay RL1 gets energized. This makes a PFC capacitor to connect across the inductive load to neutralise the lagging power factor. Similarly, port pin RC5 and RC6 can control relays RL2 and RL3 to neutralise the lagging power factor.

A 4MHz crystal oscillator is connected to pins 13 and 14 of IC1 to provide the basic clock frequency. Power-on reset is provided by the combination of resistor R1 and capaci-

tor C1. Switch S1 is used for manual reset of the microcontroller. Port pins RA0 and RA1 of IC1 receive the zero-crossing detection pulses of voltage and current, respectively.

Zero-crossing detection. The zero-crossing is the instantaneous point in an AC waveform at which there is no voltage present. In this circuit, two zero-crossing detectors are employed to get the phase angle of voltage and current. First zero-crossing detector (comprising X1, BR1, T1 and ZD1) is used for detecting the point where the voltage crosses zero in either direction of the sinusoidal AC signals (positive and negative cycles).

The mains voltage of 230V AC is stepped down by transformer X1 to deliver a secondary output of 9V. The transformer output is rectified by full-wave bridge rectifier BR1 and applied to the base of transistor T1 through resistor R6. Capacitor C4 charges to its maximum value through diode D4 and provides supply to collector of transistor T1. Zener diode ZD1 regulates the voltage to 5.1V which is suitable for microcontroller input.

When rectified output transits into zero, T1 becomes off and its collector goes high. The detected voltage pulse is applied to port pin RA0 of IC1.

Similarly, zero-crossing detection for current is done through current transformer X2, bridge rectifier BR2, transistor T2 and zener diode ZD2. The current sample is rectified by bridge rectifier BR2. When rectified output goes through zero, a pulse is generated and applied to port pin RA1 of IC1.

The time difference between the voltage and the current pulses is calculated by the program embedded in the microcontroller. The power factor value is then calculated and displayed on the LCD. By knowing the value of power factor, we can calculate the value of PFC capacitor required.

PFC capacitor value. For neutralising the inductive reactance, we need to determine the value of PFC capacitor, including its kilo-volt-ampere-reactive (kVAR) rating, which needs to be